

Question Number	Answer	Additional guidance	Mark
6(a)(i)	• vacuole	ACCEPT permanent vacuole / sap vacuole / large vacuole	(1)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	An answer that makes reference to the following point: • to {prevent / reduce} {water loss / transpiration / evaporation} through pore (1)	ACCEPT reducing diffusion of water out of stomata IGNORE water loss from cell IGNORE guard cell would have less water	(1)

Question Number	Answer	Mark
6(b)	The only correct answer is B -46.9 % <i>A is not correct because the decrease is -46.9 %</i> <i>C is not correct because the decrease is -46.9 %</i> <i>D is not correct because the decrease is -46.9 %</i>	(1)

Question Number	Answer
*6(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • there {is a lower (percentage) survival rate / are fewer plants} when there is water stress / converse • plants with water stress have a lower cellulose content / converse • plants with water stress have {a smaller root length / roots that are 4.5 cm smaller} / converse • plants with water stress have lower chlorophyll content / converse • relevant comment regarding validity of data / significant difference between the groups • lower cellulose content results in {{thinner/ weaker} cell walls / fewer cells / reduced tensile strength} / converse • {thinner cell walls / reduced calcium pectate} will result in less growth due to less support (for plants) / converse • smaller roots will absorb {less water / fewer mineral ions} in plant / converse • fewer magnesium ions results in reduced chlorophyll formation • fewer nitrates' results in reduced {amino acid / protein / chlorophyll / DNA} formation • fewer calcium ions results in reduced {calcium pectate / middle lamella} formation • less water results in {reduced photosynthesis / reduced metabolic reactions / increased wilting of plant / reduced turgidity / closure of stomata / reduced transpiration / less uptake of mineral ions} • reduced chlorophyll results in {fewer chloroplasts / less light absorption / reduced photosynthesis} • reduced photosynthetic rate results in reduced production of named photosynthetic product e.g. glucose • reduced β-glucose results in reduced cellulose formation / reduced α-glucose linked to reduced respiration • reduced {protein synthesis / photosynthetic products / mitosis} would result in decreased {growth / seed production / yield} of plants (6)

		Additional guidance
Level 0	0	No awardable content
Level 1	1-2	Limited number of the most important or relevant factors from the data/information provided are synthesised. No judgement is made.
Level 2	3-4	Some of the most important or relevant scientific factors from the data/information provided are synthesised. A limited accurate judgement is made.
Level 3	5-6	Most of the important or relevant scientific factors from the data/information provided are synthesised. A detailed and accurate judgement is made.